18/10/2026

CODE A

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 Practice Test for AIATS-01 (RM) **Time : 180 Min.**

Physics

1. (B)
2. (D)
3. (D)
4. (D)
5. (C)
6. (A)
7. (D)
8. (B)
9. (A)
10. (B)
11. (B)
12. (C)
13. (B)
14. (C)
15. (A)
16. (B)
17. (D)
18. (B)
19. (C)
20. (C)
21. (C)
22. (C)
23. (D)
24. (C)
25. (B)
26. (D)
27. (D)
28. (C)
29. (C)
30. (A)
31. (A)
32. (C)
33. (D)
34. (A)
35. (B)
36. (D)
37. (B)
38. (C)
39. (A)
40. (C)
41. (A)
42. (D)
43. (A)
44. (B)
45. (B)

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Chemistry

46. (A)
47. (B)
48. (B)
49. (D)
50. (B)
51. (B)
52. (C)
69. (C)
70. (A)
71. (B)
72. (B)
73. (B)
74. (B)
75. (D)

53. (B)
54. (B)
55. (B)
56. (C)
57. (D)
58. (A)
59. (C)
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61. (A)
62. (C)
63. (D)
64. (C)
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66. (C)
67. (C)
68. (C)
76. (D)
77. (D)
78. (D)
79. (D)
80. (C)
81. (A)
82. (C)
83. (D)
84. (C)
85. (A)
86. (A)
87. (B)
88. (D)
89. (C)
90. (C)

Botany

91. (D)
92. (A)
93. (B)
94. (C)
95. (A)
96. (A)
97. (C)
98. (B)
99. (C)
100. (C)
101. (A)
102. (B)
103. (A)
104. (B)
105. (D)
106. (A)
107. (D)
108. (C)
109. (D)
110. (D)
111. (B)
112. (C)
114. (D)
115. (A)
116. (B)
117. (D)
118. (C)
119. (B)
120. (D)
121. (B)
122. (A)
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124. (A)
125. (C)
126. (B)
127. (D)
128. (C)
129. (A)
130. (D)
131. (A)
132. (C)
133. (C)
134. (D)
135. (A)

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113. (B)

Zoology

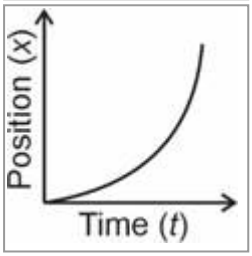
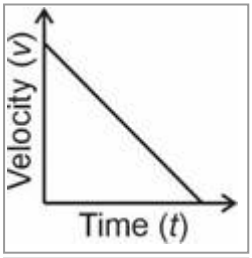
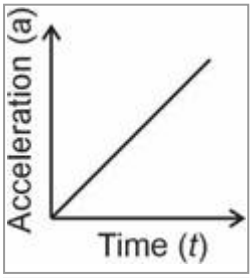
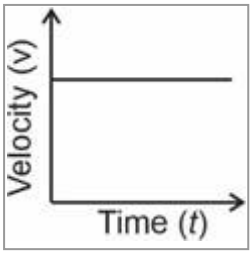
136. (C)
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176. (A)
177. (A)
178. (D)
179. (B)
180. (B)

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Hints and Solutions

Physics

(1) **Answer :** (B)
Solution:

	Column-I		Column-II
A.		(p) (r)	Non-uniform velocity Positive (non-zero) acceleration
B.		(p) (s)	Non-uniform velocity Negative (non-zero) acceleration
C.		(p) (r)	Non-uniform velocity Positive (non-zero) acceleration
D.		(q)	Uniform velocity

(2) **Answer :** (D)
Solution:

Velocity of midpoint = $\sqrt{\frac{u^2+v^2}{2}}$

(3) **Answer :** (D)
Solution:

$v_x = \frac{dx}{dt} = 5-6t$ and $v_y = \frac{dy}{dt} = 5\ \text{m/s}$

at t = 1 s, v_X = –1 m/s, v_Y = 5 m/s

$v = \sqrt{(-1)^2+5^2} = \sqrt{26}$ m/s

(4) **Answer :** (D)
Solution:

$x = R$, $y = 0$

$0 = 12R - \frac{3}{4}R^2$

R = 16 m

(5) **Answer :** (C)
Solution:

The pitch of screw gauge = $L \cdot C \cdot$ × number of divisions on circular scale.
= 0.02 × 100 = 2 mm

(6) **Answer :** (A)
Solution:

$14.45 - 13.9 = 0.55$

Result should contain only one decimal place

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$\therefore 14.45 - 13.9 = 0.6$

(7) **Answer :** (D)
Solution:
In the positive region the velocity decreases linearly (during rise) and in the negative region velocity increases linearly (during fall) and the direction is opposite to each other during rise and fall, hence fall is shown in the negative region.

(8) **Answer :** (B)
Solution:
$\text{Shortest time} = \frac{\text{d}}{\text{V}_{\text{boat}}} = \frac{1200}{2.5} = 480 \text{ s}$ = 8 minute
Direction of swimer across the rever .

(9) **Answer :** (A)
Solution:
$T = 2\pi\sqrt{\frac{L}{g}}$
$\therefore g \propto \frac{L}{T^2}$
$\frac{\Delta g}{g} \times 100 = \frac{\Delta L}{L} \times 100 + \frac{2\Delta T}{T} \times 100$
$= \left(\frac{0.1}{200}\right) \times 100 + 2 \times \frac{0.1}{400} \times 100$
= 0.1%

(10) **Answer :** (B)
Hint:
Power formula in combination of errors.
Solution:
Given $P = \frac{A^2B}{C^{2/3}D^{3/4}}$
So,
$\frac{\Delta P}{P} \times 100 = 2\frac{\Delta A}{A} \times 100 + \frac{\Delta B}{B} \times 100 + \frac{2}{3} \times \frac{\Delta C}{C} \times 100 + \frac{3}{4} \times \frac{\Delta D}{D} \times 100$
or, $\frac{\Delta P}{P} \times 100$ = 2 × 1% + 2% + $\frac{2}{3}$× 3% + $\frac{3}{4} \times 2\%$
= 2% + 2% + 2% + 1.5% = 7.5%

(11) **Answer :** (B)
Hint:
Least count of vernier calliper = value of 1 M.S.D – value of 1 V.S.D
Solution:
Value of 1 M.S.D = *c*
Value of '*a*' division of vernier scale = *bc*
Value of 1 vernier division $= \frac{bc}{a}$
L.C = $c - \frac{bc}{a} = \frac{(a-b)c}{a}$

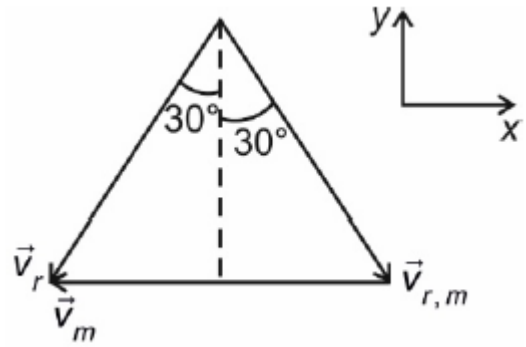
(12) **Answer :** (C)
Hint:
$T = \frac{2u\sin\theta}{g}$ and $H = \frac{u^2\sin^2\theta}{2g} = \frac{u_y^2}{2g}$
Solution:
y-component of velocity and time of flight is same for all three particles, maximum height is same.

(13) **Answer :** (B)
Hint:
$T = 2T_1 + 2T_2$
$\Delta T = 2\Delta T_1 + 2\Delta T_2$
Solution:
$T = 2 \times 40.3 + 2 \times 70.2$
= 80.6 + 140.4 = 221.0
$\Delta T = 2 \times 0.1 + 2 \times 0.2$
= 0.2 + 0.4 = 0.6
$\therefore T \pm \Delta T = (221.0 \pm 0.6)°C$

(14) **Answer :** (C)
Solution:
$T = \frac{2\pi r}{v}, a_C = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$

(15) **Answer :** (A)
Hint:
Use $\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$
Solution:

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$\vec{v}_r = \left(-20\sin 30\hat{i} - 20\cos 30^\circ \hat{j}\right)$ m/s

$= \left(-10\hat{i} - 10\sqrt{3}\hat{j}\right)$ m/s

Now $\vec{v}_m = -2\cdot\left(20\sin 30^\circ\right)\hat{i}$ m/s

$= -20\hat{i}$ m/s

(16) **Answer :** (B)
Solution:

$v_x = \frac{dx}{dt} = 7 + 8t$

$a_x = \frac{dv_x}{dt} = 8\ \mathrm{m/s^2}$

$v_y = 5 \Rightarrow a_y = 0$

$a_R = a_x = 8\ \mathrm{m/s^2}$

(17) **Answer :** (D)
Solution:

$F = \frac{2}{\sin\theta + \cos\theta}$, for *F* to be min, (sinθ + cosθ) must be max

Maximum value of (sinθ+cosθ) = $\sqrt{2}$

$F_{\min} = \frac{2}{\sqrt{2}} = \sqrt{2}$

(18) **Answer :** (B)
Solution:

|Displacement| = $\frac{1}{2} \times 2 \times 5 + 3 \times 5 = 20$ m

|Distance| = $\frac{1}{2} \times 2 \times 5 + 15 + \frac{1}{2} \times 5 \times 2 + \frac{1}{2} \times 5 \times 2 = 30\,\mathrm{m}$

(19) **Answer :** (C)
Solution:

$T = d^{\frac{a}{2}} r^{\frac{b}{2}} s^{\frac{c}{2}}$

$\left[T\right] = \left[\mathrm{ML^{-3}}\right]^{\frac{a}{2}} [L]^{\frac{b}{2}} \left[\mathrm{MT^{-2}}\right]^{\frac{c}{2}}$

$= M^{\frac{(a+c)}{2}} L^{\left(-\frac{3a}{2}+\frac{b}{2}\right)} T^{-c}$

$\Rightarrow\ -\frac{3a}{2} + \frac{b}{2} = 0$

a = 1 (given)

$\Rightarrow\ -\frac{3}{2} + \frac{b}{2} = 0\ or\ b = 3$

(20) **Answer :** (C)
Solution:

Area under *v* versus *t* graph gives displacement.

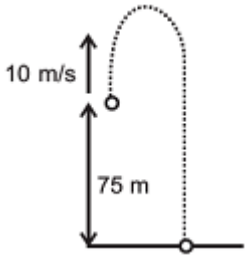
∴ Δ*x* = $\frac{1}{2} \times (4-0) \times 10 - \frac{1}{2} \times 2 \times 10$

⇒ *x*(6) – *x*(0) = 20 – 10 = 10 m

⇒ *x*(6) = 10 + *x*(0) = 0 m.

(21) **Answer :** (C)

(22) **Answer :** (C)
Solution:



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$75 = -10t + 5t^2$

$t^2 - 2t - 15 = 0$

$t = 5$ s and (-3) s since time is positive.

$\Rightarrow t = 5$ s

(23) **Answer :** (D)

(24) **Answer :** (C)

Solution:

$$\left|\vec{V}_{AB}\right|_{\text{Horizontal}} = \frac{30}{\sqrt{3}} \times \frac{\sqrt{3}}{2} + 60 \times \frac{1}{2} = 15 + 30 = 45 \text{ m/s}$$

$$d_{\text{rel}} = V_{\text{rel}} \times t \;\Rightarrow\; x = 45 \times t \;\Rightarrow\; t = \frac{x}{45}$$

(25) **Answer :** (B)

Solution:

$$t = \sqrt{\frac{2h}{g}}$$

= 4 s

$T = 8$ s

time interval between two consecutive ball

$$= \frac{T}{10}$$

= 0.8 s

(26) **Answer :** (D)

Solution:

At $t = 6$ s, v = 0 then

$t = \frac{u}{g} \Rightarrow u = gt$

$u = 10 \times 6 = 60 \text{ ms}^{-1}$

$$u = 60 \text{ ms}^{-1}$$

(27) **Answer :** (D)

Hint:

If both magnitude as well as direction of acceleration remains constant then it is uniformly accelerated motion.

Solution:

In projectile motion direction of acceleration is constant.

∴ It is uniformly accelerated motion

In uniform circular motion the direction of centripetal acceleration changes, although magnitude is constant.

∴ It is non-uniformly accelerated motion.

(28) **Answer :** (C)

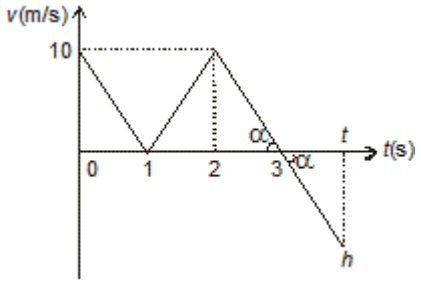
Solution:

$$v_{av} = \frac{\text{total distance}}{\text{total time}}$$

$$= \frac{S}{\frac{S}{4v_1} + \frac{3S}{4v_2}} = \frac{4v_1 v_2}{v_2 + 3v_1}$$

(29) **Answer :** (C)

Solution:



From 0 to 1 s, area $= +\frac{1}{2} \times 1 \times 10 = +5$ m

From 1 s to 3 s, area $= +\frac{1}{2} \times 2 \times 10 = +10$ m

∴ From 0 to 3 s, total area = 5 + 10 = 15 m

For zero displacement, negative area should also be 15.

Therefore, $\frac{1}{2} \times (t-3) \times h = 15$

$$\Rightarrow \frac{1}{2} \times (t-3) \times 10(t-3) = 15 \qquad \left[\because \tan\alpha = \frac{h}{t-3} = \frac{10}{1}\right]$$

$\Rightarrow (t-3)^2 = 3$

$\Rightarrow t - 3 = \pm\sqrt{3}$

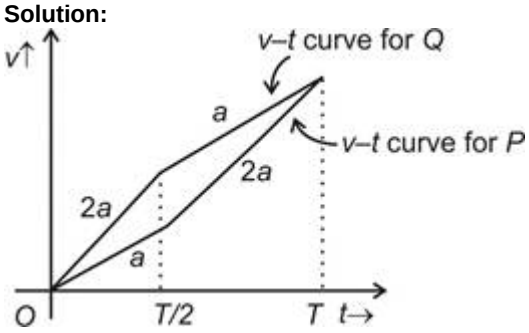
$\Rightarrow t = +\sqrt{3} + 3$

Here t must be greater than 3

∴ Answer is $(3 + \sqrt{3})s$

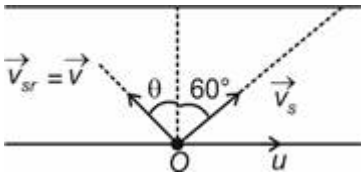
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(30) **Answer :** (A)
Solution:



Since area under *v* - *t* curve represents distance. Clearly, distance travelled by Q > distance travelled by *P*.

(31) **Answer :** (A)
Solution:



Let velocity of swimmer w.r.t river is *v* at an angle θ w.r.t. vertical as shown.

$\therefore \quad \vec{v}_s = \vec{v}_{sr} + \vec{v}_r = \left(u - v\sin\theta\right)\hat{i} + v\cos\theta\hat{j}$

$\therefore \quad \tan 60^\circ = \frac{\left(\vec{v}_s\right)_x}{\left(\vec{v}_s\right)_y} = \frac{u - v\sin\theta}{v\cos\theta}$

$\Rightarrow \quad \sqrt{3}v\cos\theta = u - v\sin\theta$

$\Rightarrow \quad v\left(\sqrt{3}\cos\theta + \sin\theta\right) = u$

$\Rightarrow \quad v = \frac{u}{(\sqrt{3}\cos\theta + \sin\theta)} = \frac{u}{2\sin(\theta + 60^\circ)}$

$\therefore$ *v* is minimum at $\theta = 30^\circ$

$\therefore \quad v_{\min} = \frac{u}{2\sin 90^\circ} = \frac{u}{2}$

(32) **Answer :** (C)
Solution:
From *X* to *Y*

$v_{av} = \frac{2v_1v_2}{v_1+v_2}$

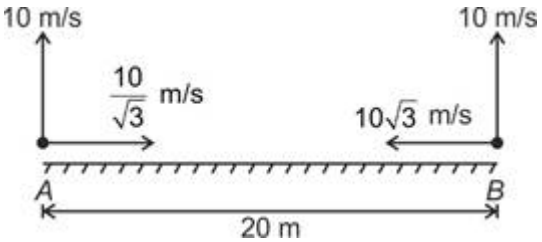
From *Y* to *X*

$v'_{av} = \frac{v_1+v_2}{2}$

$\therefore \quad \frac{v_{av}}{v'_{av}} = \frac{2v_1v_2 \times 2}{(v_1+v_2)\times(v_1+v_2)}$

$= \frac{4v_1v_2}{(v_1+v_2)^2}$

(33) **Answer :** (D)
Hint:
Relative motion in 2-D
Solution:
Resolving:



Both particles will collide ⇒ minimum separation = 0

(34) **Answer :** (A)
Hint:
$\tan\alpha = \frac{u\sin\theta - gt}{u\cos\theta}$
Solution:
Let speed of projection be *u* and angle of projection be θ. At *t* = 2 s, particle is moving horizontally. So,
u sinθ – *g* × 2 = 0
⇒ *u* sinθ = 2*g*
After 3 s, angle is 30° with horizontal.
so,
30°

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$\tan(-30^\circ) = \frac{u\sin\theta - g\times t}{u\cos\theta}$

$\Rightarrow -\frac{1}{\sqrt{3}} = \frac{2g-3g}{u\cos\theta} \Rightarrow u\cos\theta = \sqrt{3}g$

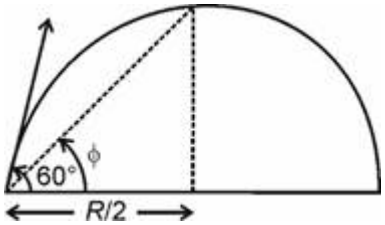
Angle of projection, $\tan\theta = \frac{u\sin\theta}{u\cos\theta} = \frac{2g}{\sqrt{3}g} = \frac{2}{\sqrt{3}}$

$\Rightarrow \theta = \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(35) **Answer :** (B)
Hint:

$\tan\phi = \frac{H}{\frac{R}{2}} = \frac{2H}{R}$

Solution:



$\tan\phi = \frac{2H}{R} = \frac{\frac{2u^2\sin^2\theta}{2g}}{\frac{u^2\times 2\sin\theta\cos\theta}{g}}$

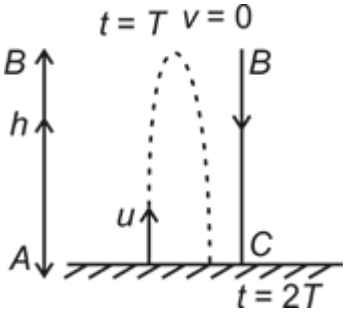
$\tan\phi = \frac{1}{2}\tan 60^\circ$

$\phi = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

(36) **Answer :** (D)
Hint:

$v = u + at$ and $v^2 = u^2 + 2as$

Solution:



At maximum height velocity becomes zero
$A \rightarrow B$
$v = u + at$
$T = \frac{u}{g}$
$B \rightarrow C$
$v = u + at$
$v = 0 + g\times\frac{u}{g}$
$v = u$ (downwards)

(37) **Answer :** (B)
Solution:

$a = 1 - t$
$\Rightarrow dV = (1-t)dt$
$\int_0^V dV = \int_0^t (1-t)dt$
$v = t - \frac{t^2}{2}$
$ds = \left(t - \frac{t^2}{2}\right)dt$
$\int ds = \int_1^3 \left(t - \frac{t^2}{2}\right)dt$
$s = \left[\frac{t^2}{2} - \frac{t^3}{6}\right]_1^3$
$s = \left[\frac{9}{2} - \frac{27}{6} - \frac{1}{2} + \frac{1}{6}\right]$
$s = \left[4 - \frac{26}{6}\right]$
$s = -\frac{1}{3}$ m

(38) **Answer :** (C)
Solution:

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If particle is moving with constant velocity then magnitude of average velocity is equal to instantaneous speed.

(39) **Answer :** (A)
Hint:
$T = \frac{2u \sin\theta}{g}$
Solution:
$T_1 = \frac{2u \sin\theta}{g} \Rightarrow \sin\theta = \frac{gT_1}{2u}$ ...(i)
$T_2 = \frac{2u \sin(90-\theta)}{g}$
$T_2 = \frac{2u \cos\theta}{g} \Rightarrow \cos\theta = \frac{gT_2}{2u}$ ...(ii)
Squaring and adding
$\sin^2\theta + \cos^2\theta = \left(\frac{gT_1}{2u}\right)^2 + \left(\frac{gT_2}{2u}\right)^2$
$u^2 = \frac{g^2}{4}\left(T_1^2 + T_2^2\right)$
$u = \frac{g}{2}\sqrt{T_1^2 + T_2^2}$

(40) **Answer :** (C)
Solution:
a = kt
$v = \int a\,dt$

$v = \int_0^t kt\,dt = \frac{kt^2}{2}$
$v \propto t^2$ (equation of upward parabola)

(41) **Answer :** (A)
Solution:
As, for freely falling bodies, relative acceleration is zero, the monkey will get hit by the bullet.

(42) **Answer :** (D)
Hint:
$\vec{v} = \frac{d\vec{r}}{dt}$; $\vec{a} = \frac{d\vec{v}}{dt}$
Solution:
$\vec{r} = \cos(3t)\hat{i} + \sin(3t)\hat{j}$
$\vec{v} = \frac{d\vec{r}}{dt} = 3\sin(3t)\hat{t} + 3\cos(3t)\hat{j}$
$\vec{a} = \frac{d\vec{v}}{dt} = -9\cos(3t)\hat{i} - 9\sin(3t)\hat{j} = -9\vec{r}$
$\therefore$ acceleration is antiparallel $\vec{r}$

(43) **Answer :** (A)
Hint:
In uniform circular motion, speed $\left(\left|\vec{v}\right|\right)$ is constant, but velocity $\left(\vec{v}\right)$ is variable due to change in its direction.
Solution:
Speed is constant $\therefore \frac{d\left|\vec{v}\right|}{dt} = 0$
Velocity is variable $\therefore \frac{dv}{dt} \neq 0$
Acceleration vector is also variable $\therefore \frac{d^2\vec{v}}{dt^2} \neq 0$

(44) **Answer :** (B)
Solution:
From *O* to *B*, graph is concave down
$\therefore$ acceleration is negative.

(45) **Answer :** (B)
Solution:
Mole is unit of amount of substance.

Chemistry

(46) **Answer :** (A)
Solution:
$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
$\therefore \frac{n_{Fe_2O_3}}{1} = \frac{n_{CO}}{3} \Rightarrow n_{CO} = 3 \times n_{Fe_2O_3}$
$2C + O_2 \rightarrow 2CO$

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$\therefore \ \frac{n_{O_2}}{1} = \frac{n_{CO}}{2} \ \therefore \ n_{O_2} = \frac{3 \times n_{Fe_2O_3}}{2}$

$\therefore$ Volume of O_2 required at STP

$= \frac{3}{2} \times \frac{1}{160} \times 22400 = 210$ mL

(47) **Answer :** (B)
Solution:
Molar mass of Mg = 24 g/mole
0.04 g of Mg is present in 100 g of biomolecule
24 g of Mg is present in $\frac{100}{6.04} \times 24 = 60000$

(48) **Answer :** (B)
Solution:
Number of gram atoms = $\frac{w}{M}$
$= \frac{d \times V}{M} = \frac{13.6 \times 100}{200} = 6.8$

(49) **Answer :** (D)
Solution:
$Fe(CN)_2 + 4KCN \rightarrow K_4[Fe(CN)_6]$
$\therefore$ 3.2 mole of KCN are required

(50) **Answer :** (B)
Solution:
$H_2SO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + H_2O + CO_2$
98 g 106 g
80 g H_2SO_4 is present in 100 g of impure sample
80 g H_2SO_4 is present $\frac{100 \times 98}{80} = 122.5$ g

(51) **Answer :** (B)
Solution:
$m = \frac{x_{solute}}{1 - x_{solute}} \times \frac{1000}{M_{solvent}}$

(52) **Answer :** (C)
Solution:
At S.T.P
$\Rightarrow \ \frac{V(L)}{22.4} = \frac{Mass}{Molar \ mass}$
$\Rightarrow \ \frac{5.6}{22.4} = \frac{16}{Molar \ mass}$
$\Rightarrow$ Molar mass = 64.
Now,
Number of atoms $= \frac{64}{64} \times N_A \times 3 = 3 \ N_A$

(53) **Answer :** (B)
Hint:
$\text{Molality} = \frac{\text{Mole of glucose}}{\text{Mass of water(kg)}}$
Solution:
Mass of glucose = 36 g
Mass of water = (100 – 36) = 64 g
$\text{Molality} = \frac{36/180}{\frac{64}{1000}}$
$= \frac{36}{180} \times \frac{1000}{64}$
= 3.125 m

(54) **Answer :** (B)
Hint:
% by mass of H_2O
$= \frac{\text{Mass of } H_2O}{\text{Mass of compound}} \times 100$
Solution:
$63 = \frac{x \times 18}{18x + 106} \times 100$
$\therefore \ \boxed{x = 10}$

(55) **Answer :** (B)
Hint:
% of an element $= \frac{wt. of \ element}{wt. of \ compound} \times 100$
Solution:
For YO: % of oxygen = $\frac{16}{M + 16} \times 100$

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$40 = \frac{1600}{M+16} \quad \therefore \; M = 24$

For second oxide: Let Y_yO_x

% of oxygen = $\frac{16x}{16x+24y} \times 100$

$20 = \frac{1600x}{16x+24y}$

$\frac{x}{y} = \frac{3}{8}$

$\therefore \; Y_yO_x = Y_8O_3$

(56) **Answer :** (C)
Solution:
A tetra-atomic gas molecule contains 4 atoms
1 mole tetra-atomic gas = 4 moles of atoms
0.25 mole tetra-atomic gas = 4 × 0.25 = 1 mole atom
= N_A atoms

(57) **Answer :** (D)
Solution:
Average atomic mass = $\frac{100\times80+99\times16+101\times4}{100}$
$= \frac{8000+1584+404}{100} \approx 99.88$

(58) **Answer :** (A)
Solution:
$\text{Mass of } CH_4 = \frac{16\times3.01\times10^{23}}{6.02\times10^{23}} = 8g$

(59) **Answer :** (C)
Solution:
Mass of 1 atom = 1.8×10^{-22} g
Mass of 6.022×10^{23} atoms
= $1.8 \times 10^{-22} \times 6.022 \times 10^{23}$ = 108.39 g
= Atomic mass

(60) **Answer :** (A)

(61) **Answer :** (A)

(62) **Answer :** (C)
Solution:
$x = \frac{(x+2)y+(x-1)(1-y)}{1}$
As y is fraction abundance of mass number (x + 2) and (1 – y) is fraction abundance of mass number (x – 1)
3y – 1 = 0
3y = 1
$y = \frac{1}{3} = 33.3\%$

(63) **Answer :** (D)
Hint:
$FeO + \frac{1}{4}O_2 \rightarrow \frac{1}{2}Fe_2O_3$
Solution:
$\underset{1 \text{ mole}\ (72g)}{FeO} + \underset{0.25 \text{ mole}\ (8g)}{\frac{1}{4}O_2} \rightarrow \frac{1}{2}Fe_2O_3$

% increase in wt. $= \frac{8}{72} \times 100 = 11.11\ \%$

(64) **Answer :** (C)
Hint:

$Cr_2O_7^{2-}$ = meq of $C_2O_4^{2-}$

Solution:
$\underset{n_f=6}{Cr_2O_7^{2-}} + \underset{n_f=2}{C_2O_4^{2-}} \rightarrow Cr^{3+} + CO_2$

$N_1V_1 = N_2V_2$
$n_1M_1V_1 = n_2M_2V_2 \Rightarrow 6 \times 0.2 \times V_1 = 2 \times 0.3 \times 50$ V_1 = 25 ml

(65) **Answer :** (B)
Solution:
Basicity of oxalic acid is 2
∴ E = $\frac{126}{2}$ = 63

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(66) **Answer :** (C)
Solution:
Mole of $H_2O = \frac{9}{18} = 0.5$
Number of neutron = 0.5 × 8 × N_A
= 4 N_A

(67) **Answer :** (C)
Solution:
°F = $\frac{9}{5}$ (°C) + 32
= $\frac{9}{5}$ × 65 + 32
= 117 + 32
= 149

(68) **Answer :** (C)
Hint:
PE = –2KE
Solution:
Let kinetic energy be initially E_1, then $PE_1 = -2E_1$
New kinetic energy will be
$E_2 = E_1 - \frac{E_1}{9} = \frac{8E_1}{9}$
New potential energy $PE_2 = -\frac{16E_1}{9}$
Change in potential energy = $\left(\frac{-16E_1}{9}\right) - \left(-2E_1\right)$
$= \frac{2E_1}{9}$

(69) **Answer :** (C)
Solution:
n = 2
l = 0, 1
Number of degenerate orbitals = 4

(70) **Answer :** (A)
Solution:
$h\upsilon = w_0 + KE$
$\frac{hc}{\lambda} = w_0 + 2$ ...(1)
$\frac{4hc}{\lambda} = w_0 + 10$ ...(2)
$4(w_0 + 2) = w_0 + 10$
$3w_0 = 2$
$w_0 = \frac{2}{3}$ eV.

(71) **Answer :** (B)
Solution:
Second line of Balmer series (He^+)
$\frac{1}{\lambda} = R \times \left(4\right) \left[\frac{1}{4} - \frac{1}{16}\right] = 4R \times \frac{3}{16} = \frac{3R}{4}$
$\lambda = \frac{4}{3R}$
First line of Lyman series (H–atom)
$\frac{1}{\lambda} = R \times \left(1\right) \left[\frac{1}{1} - \frac{1}{4}\right] = \frac{3}{4}R$
$\lambda = \frac{4}{3R}$

(72) **Answer :** (B)
Solution:
$\Delta x \Delta p = \frac{h}{4\pi}$
$\Delta x \cdot m\Delta V = \frac{h}{4\pi}$
$\Delta x = 2m\Delta v$
$\therefore \ 2m\Delta v . m\Delta v = \frac{h}{4\pi}$
$\Delta v^2 = \frac{h}{8\pi m^2}$
$\Delta v = \sqrt{\frac{h}{8\pi m^2}} = \frac{1}{2m}\sqrt{\frac{h}{2\pi}} = \frac{1}{2m}\sqrt{\hbar}$

(73) **Answer :** (B)
Solution:

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$\frac{1}{2}mv^2 = qV = v = \sqrt{\frac{2qv}{m}}$

$\lambda = \frac{h}{mv} = \frac{h}{m\sqrt{\frac{2qV}{m}}} = \frac{h}{\sqrt{2qVm}}$

$\lambda = \frac{h}{\sqrt{2qm}} \cdot \frac{1}{\sqrt{V}}$

y m x

(Slope)

Since slope of particle B is greater than slop of particle A therefore $m_B < m_A$

(74) **Answer :** (B)

(75) **Answer :** (D)

Solution:

$(E_H)_n = \frac{(E_2)_n}{Z^2}$

$(E_H)_1 = \frac{\left(E_{H_2}\right)_1}{2^2} = \frac{-871.6}{4} \times 10^{-20}$ J

= –217.9 × 10^{-20} J

(76) **Answer :** (D)

Hint:

Potential energy = 2 × $\left(-13.6\frac{Z^2}{n^2}\right) eV$

Solution:

$\frac{(PE)_{He^+}}{(PE)_{Be^{3+}}} = \frac{2\times -13.6\times \frac{(2)^2}{(4)^2}}{2\times -13.6\times \frac{(4)^2}{(2)^2}}$

$= \frac{(2)^2\times(2)^2}{(4)^2\times(4)^2} = \frac{1}{16}$

(77) **Answer :** (D)

Hint:

Orbital angular momentum = $\sqrt{\ell(\ell+1)}\frac{h}{2\pi}$

Solution:

$\because \quad \ell = 0$ for *s*-orbital

$\therefore$ Orbital angular momentum for *s*-electron = 0

(78) **Answer :** (D)

Hint:

$h\nu = h\nu_0 + \frac{1}{2}m_e v^2$

Solution:

If the frequency of light used is greater than the threshold frequency only then electrons are ejected from metal.

(79) **Answer :** (D)

Solution:

For a given l value, m can be

–l, –(l – 1) ……0, 1 ….. (l – 1), l

–l to +l including zero.

(80) **Answer :** (C)

Hint:

Radial node = n – l – 1

Angular node = l

Solution:

For 4*d* orbital

Radial node = 4 – 2 – 1 = 1

Angular nodes = l = 2

(81) **Answer :** (A)

Solution:

Correct order of energy of 2*s* orbital of the given elements is

$E_{2S}(H) > E_{2S}(Li) > E_{2S}(Na) > E_{2S}(K)$

(82) **Answer :** (C)

Solution:

Since $\Delta E = \frac{hc}{\lambda}$

$\therefore$ Shortest l $\Rightarrow \Delta E$ = Maximum $\Rightarrow$ last line

So, for Lyman series, $n_2 = \infty \rightarrow n_1 = 1$

$\frac{1}{\lambda} = R_H(1)^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$

$\frac{1}{x} = R_H$

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Also, longest λ ⇒ ΔE = minimum ⇒ first line
So, Balmer series $n_2 = 3 \rightarrow n_1 = 2$

$\frac{1}{\lambda} = R_H(1)^2 \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

$= R_H \times 4 \times \frac{5}{36}$

$\lambda = \frac{9}{5R_H} = \frac{9x}{5}$

(83) **Answer :** (D)
Hint:
Write electronic configuration of given ions
Solution:
Cr^{2+} (22) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^4$
Co^{2+}(25) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^7$
Fe^{2+}(24) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$
Mn^{2+}(23) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$
Mn^{2+} has five unpaired electrons.

(84) **Answer :** (C)
Solution:
$E = \frac{nhc}{\lambda}$

$\frac{200 \times 10}{100} = \frac{n \times 6.6 \times 10^{-34} \times 3 \times 10^8}{1.4 \times 10^{-7}}$

$n = \frac{20 \times 1.4 \times 10^{-7}}{6.6 \times 10^{-34} \times 3 \times 10^8}$

$n = 1.4 \times 10^{19}$ photons s^{-1}

(85) **Answer :** (A)
Solution:
I ⇒ 4*p*
II ⇒ 4*s*
III ⇒ 3*d*
IV ⇒ 3*p*
energy is decided according to (n+l) rule higher the value of (n+l) higher is the energy. if (n+l) is same then higher the value of n higher is the energy

(86) **Answer :** (A)
Solution: 3d z^2 orbital does not have any nodal planes, but rather has two nodal cones

(87) **Answer :** (B)

(88) **Answer :** (D)
Solution:
$\frac{(r_4)_{He^+}}{(r_2)_{Be^{3+}}} = \frac{4^2 \times 4}{2 \times 2^2} = 8 : 1$

(89) **Answer :** (C)
Solution:
For 1st orbital, the probability density is maximum at nucleus.

(90) **Answer :** (C)
Hint:
$Cr[24]\ :\ 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
Solution:
Orbitals with 'n + *l* = 4' are ⇒ 3*p* and 4*s* and total electrons present in these are 7.

Botany

(91) **Answer :** (D)
Solution:
Human – Species; Mammal – Class
Animal – Kingdom.
Human, mammals and animals represent different taxa at different levels.

(92) **Answer :** (A)
Solution: Animal cells have centrioles which are absent in higher plant cells. Plant cells possess cell walls, plastids, and a large central vacuole which are absent in animal cells.

(93) **Answer :** (B)
Solution:
A meiocyte
– In interphase (before S phase) has 2n chromosomes (24), chromatids (24), 2C DNA (20 pg).

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– After S phase has 2n chromosomes (24), chromatids (48), 4C DNA (40 pg).
– After meiosis-I daughter cells have n chromosomes, chromatids (24) and 2C DNA amount (20 pg).

(94) **Answer :** (C)
Solution:
Outer membrane of chloroplast is relatively more permeable and has porins.
Grana is the active site for light reaction.
Stroma is the site for dark reaction or fixation of CO_2 into sugar.

(95) **Answer :** (A)
Hint:
Middle lamella has calcium and magnesium pectate.
Solution:
Middle lamella has mainly calcium pectate. Trans face of Golgi complex is maturing face from where large vacuoles are discharged.

(96) **Answer :** (A)
Solution:
Nuclear division and cytokinesis occur twice in meiosis.

(97) **Answer :** (C)
Hint:
Chlorophyll traps light energy for photosynthesis.
Solution:
Ribosomes are the site of protein synthesis. Chromoplasts contain carotenoid pigments and amyloplast stores starch.

(98) **Answer :** (B)
Solution:
Its parental cell had sixteen chromosomes which form eight bivalent or tetrad during pachytene of prophase-I during meiosis-I.

(99) **Answer :** (C)
Solution:
Musca belongs to order Diptera.

(100) **Answer :** (C)
Solution:
Plants belong to family Solanaceae are dicots whereas that of Poaceae are monocots.

(101) **Answer :** (A)
Solution:
Bacteria can be classified into two groups on the basis of the difference in the cell envelopes and the manner in which they respond to the Gram stain.

(102) **Answer :** (B)
Solution:
Lipids are arranged within the membrane with the polar head towards the outer sides and the non-polar tails towards the inner side.

(103) **Answer :** (A)
Hint:
A bivalent is set of homologous chromosomes.
Solution:
Cell having 5 bivalents means
(1) It is a diploid cell
(2) It has 10 chromosomes in total (univalents)
(3) It has 5 sets of homologous chromosomes
(4) It has 5 tetrads (a group of four chromatids)

(104) **Answer :** (B)
Solution:
In anaphase I, the homologous pair of chromosome separates, but each chromosome still contains both sister chromatids as the centromere does not split in this phase.

(105) **Answer :** (D)
Solution:
In telophase, reappearance of nucleolus, golgi complex and E.R. occur. Chromosomes lose their identity as discrete elements.

(106) **Answer :** (A)
Solution:
Shape of WBCs is amoeboid, RBCs are round and biconcave, nerve cell is branched and long and mesophyll cells are round and oval.

(107) **Answer :** (D)
Solution:
80S ribosomes consist of 60S and 40S subunits.

(108) **Answer :** (C)
Solution:

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Leucoplast is colourless plastid. The orange colour of carrot is due to chromoplasts.

(109) Answer : (D)
Solution:
Nucleolus is a site of active ribosomal RNA synthesis. Larger and more numerous nucleoli are present in the cells actively carrying out protein synthesis.

(110) Answer : (D)
Solution:
Glycocalyx could be loose sheath called the slime layer in some while in others, it may be thick and tough called the capsule.

(111) Answer : (B)
Solution:
Reserve material in prokaryotic cells are stored in the cytoplasm, in the form of inclusion bodies.

(112) Answer : (C)
Hint:
A bivalent is a pair of homologous chromosomes.
12 bivalents = 24 chromosomes. Hence in gamete it will be 12 chromosomes.
Solution:
One chromosome has one chromatid in gamete. So, total 12 chromosomes & 12 chromatids.

(113) Answer : (B)
Solution:
Convolvulaceae and Solanaceae are included in order Polymoniales mainly based on the floral characters.

(114) Answer : (D)
Solution:
Telophase is last phase of karyokinesis. Compaction of chromosomes completes in metaphase.

(115) Answer : (A)
Hint:
Crossing over occurs only once during reductional division.
Solution:
Recombination or crossing over between homologous chromosomes occur once in meiosis. Duplication of centrioles, karyokinesis and metaphasic plate formation occur twice in the whole meiosis.

(116) Answer : (B)
Solution:
Dogs and cats belong to two different genera.

(117) Answer : (D)
Solution:
Among family, genus, class and phylum; phylum will have least number of common characters.

(118) Answer : (C)
Solution:
Diptera and Insecta are Order and Class, respectively.

(119) Answer : (B)
Hint:
Chromoplasts are formed either from leucoplast or chloroplast.
Solution:
Change of colour from green to reddish during ripening of tomato and chilli is due to transformation of chloroplast to chromoplast.

(120) Answer : (D)
Hint:
Number of meiotic divisions to produce 'n' numbers of angiospermic seeds = $n + \frac{n}{4}$
Solution:
To produce 160 angiospermic seeds, number of meiotic divisions required are; $160 + \frac{160}{4} = 200$

(121) Answer : (B)
Hint:
Recombinase enzyme catalyses the process occurring in pachytene stage.
Solution:
Crossing over is enzyme mediated process and it produces recombinants.

(122) Answer : (A)
Solution:
Crossing over results into the variations and genetic recombinations, as it occurs between the non-sister chromatids of homologous chromosome.

(123) Answer : (B)
Solution:
During anaphase II, the sister chromatids separate.

(124) Answer : (A)
Solution:

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Dissolution of synaptonemal complex – Diplotene
Formation of recombination nodule – Pachytene
Assembling of meiotic spindle at the poles – Diakinesis.

(125) **Answer :** (C)
Solution:
Centriole has an organisation like the cartwheel. It is surrounded by amorphous pericentriolar materials.

(126) **Answer :** (B)
Hint:
Mitochondria are semi-autonomous cell organelles.
Solution:
Matrix contains single circular dsDNA (with high G ≡ C contents), a few RNA molecules, 70S ribosomes and the components required for synthesis of protein. It contains enzymes for TCA cycle.

(127) **Answer :** (D)
Solution:
Centriole gives cartwheel appearance.

(128) **Answer :** (C)
Solution:
The vacuole is bound by a single membrane called tonoplast.

(129) **Answer :** (A)
Solution:
A prokaryotic cell does not have membrane-bound cell organelles.

(130) **Answer :** (D)
Solution:
Lysosomes are very rich in almost all types of hydrolytic enzymes.

(131) **Answer :** (A)
Solution:
The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure.

(132) **Answer :** (C)
Solution:
Rudolf Virchow explained **"*Omnis cellula e cellula* ".**

(133) **Answer :** (C)
Solution:
Mesosome helps in respiration , secretion processes, DNA replication and cell wall formation.

(134) **Answer :** (D)

(135) **Answer :** (A)
Solution:
Cisternae in Golgi apparatus are interconnected.

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Zoology

(136) **Answer :** (C)
Solution:
Prosthetic group and co-enzyme both are organic co-factors.

(137) **Answer :** (A)
Hint:
Hydroxyl group is present as a side chain in alcoholic amino acids.
Solution:
The given amino acid structure is that of alanine as methyl ($-CH_3$) is the side chain. Palmitic acid is a 16–C saturated fatty acid.

(138) **Answer :** (C)
Hint:
Free amino group is present in first amino acid.
Solution:
In a long peptide chain, the first amino acid contains free amino group, hence it is also called the N-terminal and the last amino acid contains free carboxyl group, hence it is also called C-terminal amino acid. The first amino acid is present at left end, called N-terminal and the last amino acid is present at right end, called C-terminal.

(139) **Answer :** (B)
Solution:
Pseudostratified epithelium lines larger bronchi.

(140) **Answer :** (B)
Solution:

In a nucleic acid, a phosphate moiety links the 3'-carbon of one sugar of one nucleotide to the 5'-carbon of the sugar of the succeeding nucleotide.

(141) Answer : (B)
Solution:

Ciliated epithelium	– Inner lining of oviduct
Squamous epithelium	– Walls of blood vessels
Cuboidal epithelium	– PCT of nephron
Compound epithelium	– Moist surface of buccal cavity

(142) Answer : (D)
Solution:
In competitive inhibition, substrate and inhibitor compete for same catalytic site but only one can bind to it at a time.

(143) Answer : (D)
Solution:
In the presence of a competitive inhibitor, the Km value increases and the Vmax remains constant.

(144) Answer : (B)
Solution:
Thiamine is vitamin B_1; Inulin is a storage polysaccharide; hyaluronidase is an enzyme which is heteropolymer.

(145) Answer : (C)
Solution:
Oligodendrocytes form myelin sheath in CNS. Astrocytes form blood brain barrier while Schwann cells form myelin sheath in PNS.

(146) Answer : (C)
Solution:
In DNA, deoxyadenylic acid is present.

(147) Answer : (A)
Solution:
A number of enzymes require metal ions for their activity, such as zinc acts as a co-factor for carboxypeptidase. Vitamins are the essential chemical components of many co-enzymes, *e.g.*, co-enzyme nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin.

(148) Answer : (A)
Hint:
Glycogen can be stored in human liver.
Solution:
Chitin is a structural homopolysaccharide, made up of of N-acetylglucosamine units. Cellulose is a structural homopolysaccharide that does not give positive iodine test.

(149) Answer : (A)
Hint:
Tripeptide consists of three amino acids.
Solution:
First amino acid of chain is serine which is towards N-terminal.
Cytosine is the nitrogenous base.
Given structure is in primary configuration.

(150) Answer : (B)
Solution:
The given structures are palmitic acid (saturated fatty acid), cholesterol (a sterol), uracil (a pyrimidine) and a glucose (hexose sugar), respectively.

(151) Answer : (A)
Hint:
Brush bordered epithelium (BBE) increases surface area for absorption.
Solution:
Brush bordered epithelium increases reabsorption in PCT of nephron and absorption in intestine.

(152) Answer : (C)
Solution:
When enzyme is absent there will be more time taken for a reaction to start and hence, there will be less product.

(153) Answer : (D)
Solution:
$-NH_2$ group is free in the **first** amino acid: N-terminal.
–COOH group is free in the **last** amino acid: C-terminal.

(154) Answer : (B)
Hint:
Prosthetic groups are tightly bound to the apoenzyme
Solution:
Coenzyme nicotinamide adenine dinucleotide contain the vitamin niacin. Zinc is a co-factor for the proteolytic enzyme carboxypeptidase.

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(155) Answer : (D)
Hint:
Structure of compound epithelium.
Solution:
Compound epithelium covers the dry surface of the skin, the moist surface of buccal cavity, pharynx, inner lining of ducts of salivary glands and of pancreatic ducts. The main function of compound epithelium is to provide protection against chemical and mechanical stresses.

(156) Answer : (C)
Solution:
Osteoblast – Bone forming cells
Osteocytes – Bone maintaining cells
Osteoclasts – Resorption of bone
Chondroblasts – Cartilage forming cells
Chondrocytes – Cartilage maintaining cells
Chondroclasts – Resorption of cartilage

(157) Answer : (B)
Solution:
According to Chargaff's rule, A = T and G = C
So, if A is 20% then T is 20%, G and C will be 30% each.

(158) Answer : (A)
Solution:
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(159) Answer : (A)
Solution:
Cytidylic acid is present in RNA and DNA whereas uridylic acid is present only in RNA.

(160) Answer : (D)
Hint:
Prostaglandins are derived from 20C unsaturated fatty acid.
Solution:
Prostaglandins are paracrine hormones and are derived lipids obtained from arachidonic acid. Arachidic acid is 20C saturated fatty acid.

(161) Answer : (B)
Solution:
Basement membrane is a non-cellular layer.
There is little intercellular material between the cells.
Buccal cavity, pharynx and urinary bladder are composed of compound epithelium

(162) Answer : (C)
Solution:
Competitive inhibitors increase the value of the Michaelis constant (K_m), but do not modify the maximum velocity (V_{max}) of the enzyme.

(163) Answer : (B)
Solution:
Trypsin is a proteolytic enzyme. Antibodies are proteinaceous compounds which help to fight against infectious agents. GLUT-4 is a proteinaceous transporter for glucose. Insulin is a hypoglycemic proteinaceous hormone. Collagen is the most abundant protein in the animal world.

(164) Answer : (C)
Solution:
Dense regular connective tissues : Bone and cartilage are specialised connective tissues.

(165) Answer : (D)
Hint:
Each complete turn of B-DNA has 10 base pairs.
Solution:
Pitch of B-DNA = 34 Å
1 nm = 10 Å
So, 68 nm = 68 × 10 = 680 Å
34 Å length of B-DNA contain 10 base pairs
So, base pair in 680 Å length = $\frac{680 \times 10}{34}$ = 200
So, total number of nucleotides = 200 × 2
= 400 nucleotides

(166) Answer : (C)
Solution:
Simple carbohydrate cannot be hydrolysed further into smaller components.

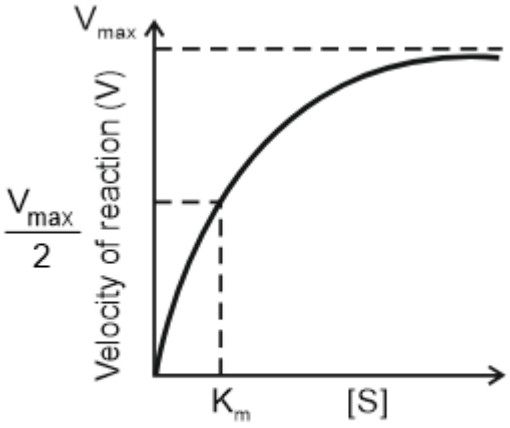
(167) Answer : (D)
Hint:
Vitamins are essential component of coenzymes.
Solution:

Haem is the prosthetic group for the enzymes named catalase and peroxidase. Niacin is present in co-enzymes NAD and NADP. Zinc is the co-factor for enzymes carbonic anhydrase and carboxypeptidase.

(168) Answer : (D)
Solution:
Dipeptides have two amino acids, glycosidic bond formation is accompanied by loss of water. Lipids are not polymers.

(169) Answer : (C)
Solution:
The enzymes isolated from thermophilic bacteria do not loose their activity at higher temperature.

(170) Answer : (B)
Solution:



Effect of change in concentration on enzyme activity

(171) Answer : (B)
Solution:
» Given structure is uridine [nucleoside] that is formed by joining of ribose to uracil.
» Becomes nucleotide by adding one phosphate.

(172) Answer : (A)
Solution:
Epithelium can be simple or compound.
Simple epithelium is composed of a single layer of cells, whereas compound epithelium consists of two or more cell layers. Epithelium is avascular.

(173) Answer : (A)
Hint:
Neurotransmitters are present at synapses.
Solution:
Gap junctions provide cytoplasmic channels to facilitate communication between adjacent cells. Adhering junctions aid to cement adjacent cells to form a sheet.
Tight junctions provide a barrier which prevent leakage of fluid across epithelial cells.

(174) Answer : (D)
Solution:
GLUT-4 is a protein, that enables glucose transport into cells.

(175) Answer : (A)
Hint:
Compare energy of product that is higher than energy of substrate.
Solution:
Enzymes decrease energy of activation so energy of transition is lower in presence of enzyme in comparison to non-enzymatic reaction. If energy of product is more than energy of substrate, reaction is endothermic and vice versa.

(176) Answer : (A)
Solution:
According to Chargaff's rule, A = T and G = C, thus total number of nucleotides is 400.
Total number of H-bonds between A and T is 100 x2 = 200
Total number of H-bonds between G and C is 100 x3 = 300
Thus, total number of H-bonds = 500

(177) Answer : (A)
Hint:
Brush bordered epithelium (BBE) increases surface area for absorption.
Solution:
Brush bordered epithelium increases reabsorption in PCT of nephron and absorption in intestine.

(178) Answer : (D)
Hint:
Select a cell which forms myelin sheath in PNS.

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Solution:
Astrocytes/Macrocytes are large glial cells with a number of protoplasmic processes. They form maximum number of glial cells. They help in repair of nervous tissue and form blood brain barrier. Schwann cells form myelin sheath and neurilemma around an axon in PNS.

(179) **Answer :** (B)
Solution:
All are chemically proteins.

(180) **Answer :** (B)
Solution:
Non-competitive inhibitor gets attached to the site other than active site and change the configuration of active site. Rate of reaction decreases but the affinity of enzyme for substrate remains same.

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